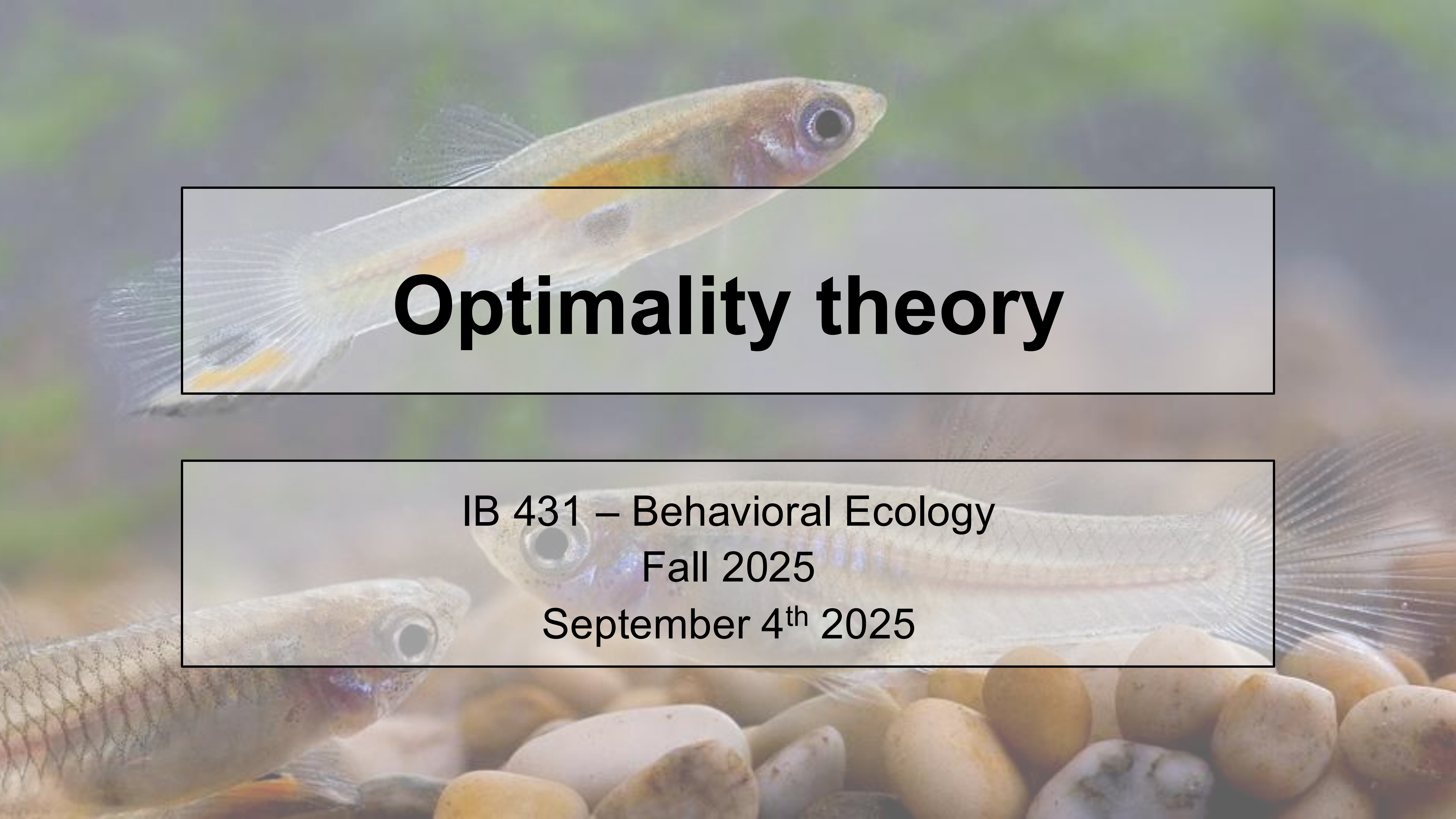


Optimality theory

IB 431 – Behavioral Ecology
Fall 2025
September 4th 2025

Outline

I. Optimality theory and examples of optimality models
II. Work through optimality problem
III. Go over proposal and first proposal assignment
IV. Discuss Abrahams and Dill 1989

Optimality theory

- Aims to model natural selection as a function of the costs and benefits of different behaviors and choices
- Selection acts as an “optimizing” agent

Optimality theory

- Aims to model natural selection as a function of the costs and benefits of different behaviors and choices
- Selection acts as an "optimizing" agent
- **Reduces complex problems (i.e., Where to forage? Who to mate with? Which food to eat? What group size to be in?) to simple mathematical terms**

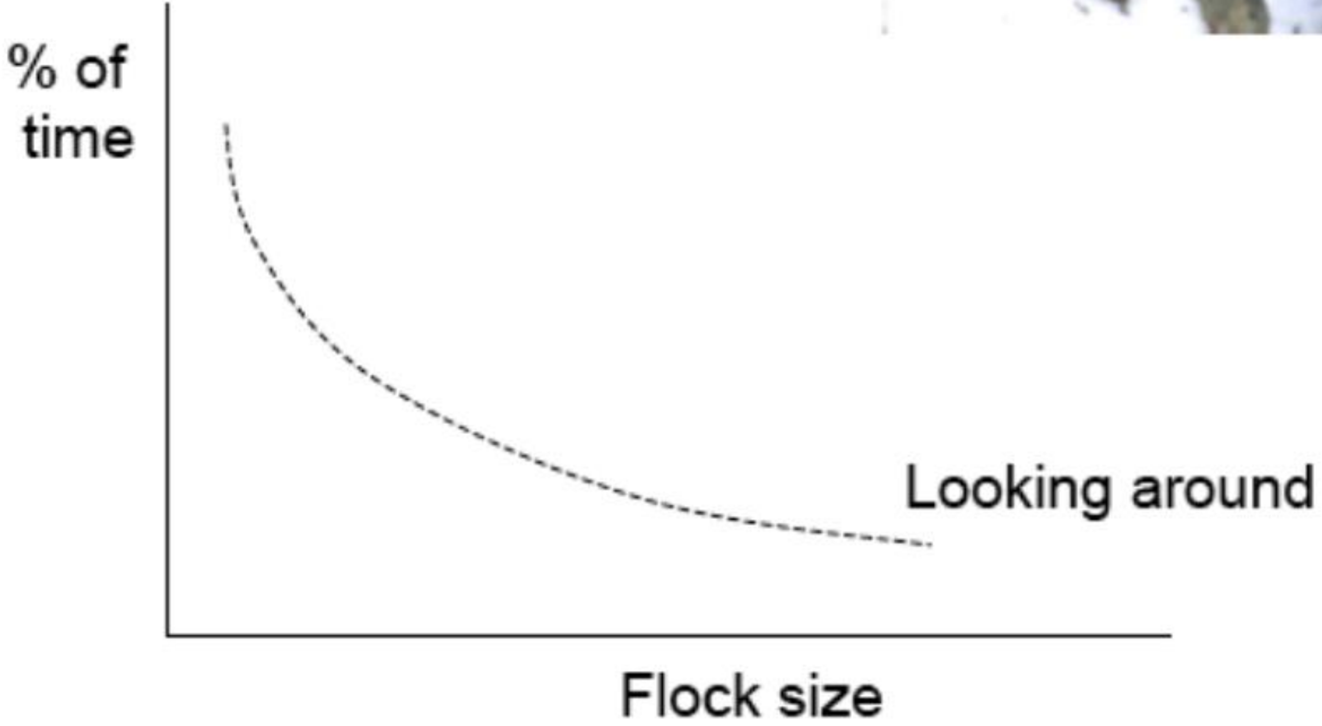
How do we build an optimality model?

1. Identify the problem – Finding food? Mates?
2. Choose a currency
    - Ideally, a measure of fitness (# of offspring, clutch size, etc.)
    - Otherwise, can use a fitness correlate (energy gain, time, rate of fertilization, risk of predation, etc.)
3. Assess situation / any constraints or alternative choices
4. Quantify costs and benefits of available alternatives (using your currency as a measure)

Example: Optimal group size in birds

1. Identify the problem:
   What group size to forage in?
2. Choose a currency: Time
3. Assess situation/alternatives:
   Bird must spend time scanning for predators and fighting with other birds
4. Quantify costs and benefits of available alternatives:
   Smaller flocks need to check for predators more but fight less; larger flocks need to check for predators less but fight more

What is the best size group to forage in?



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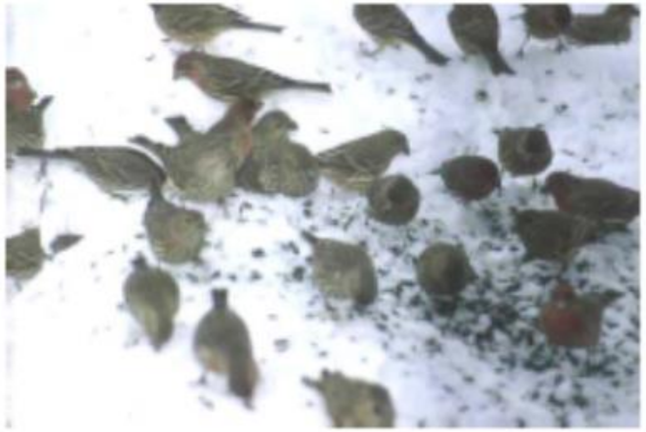

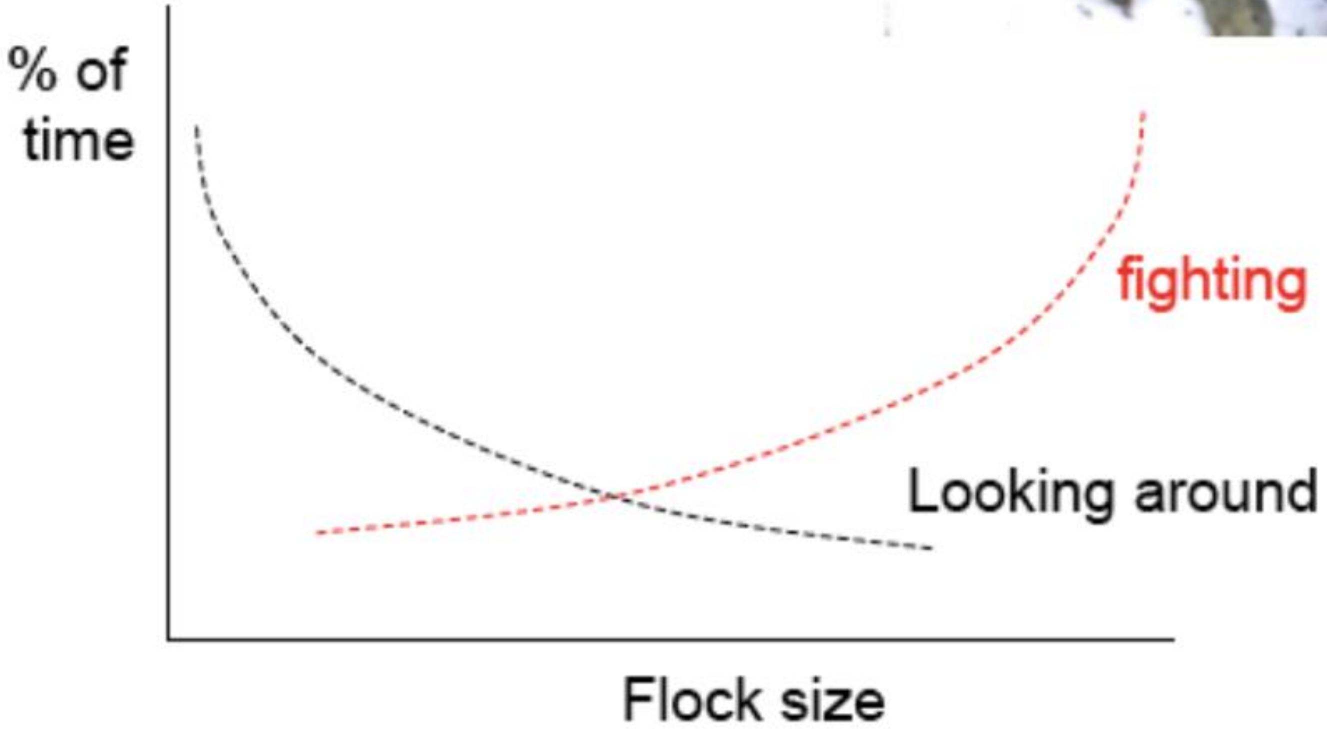

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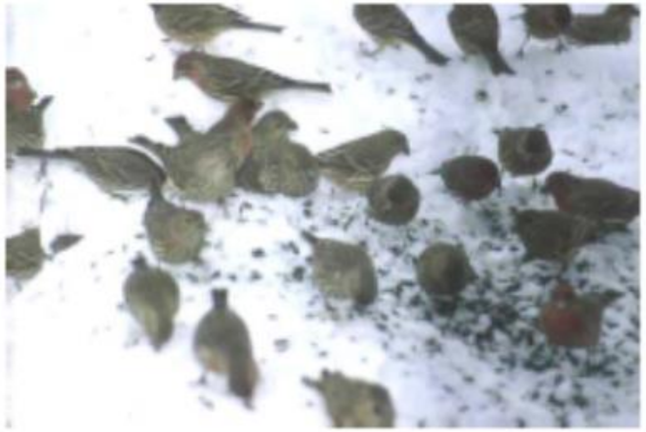
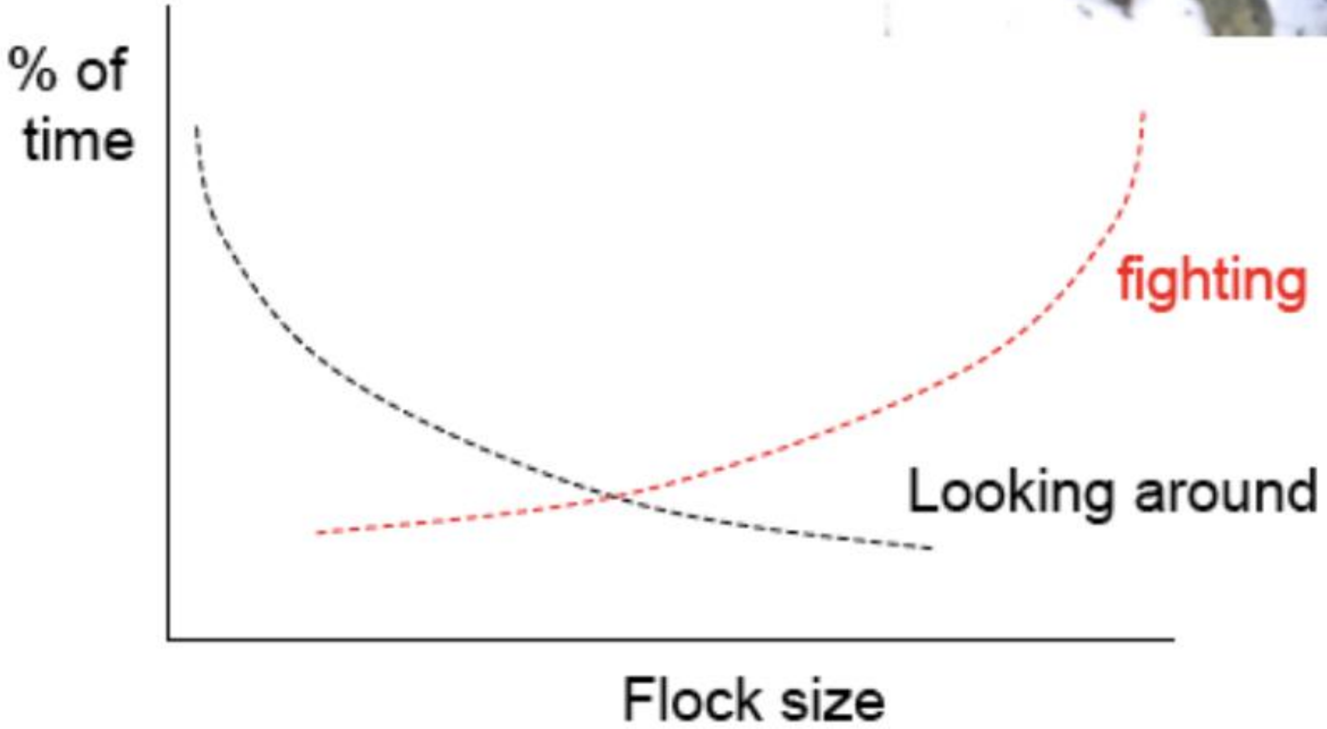
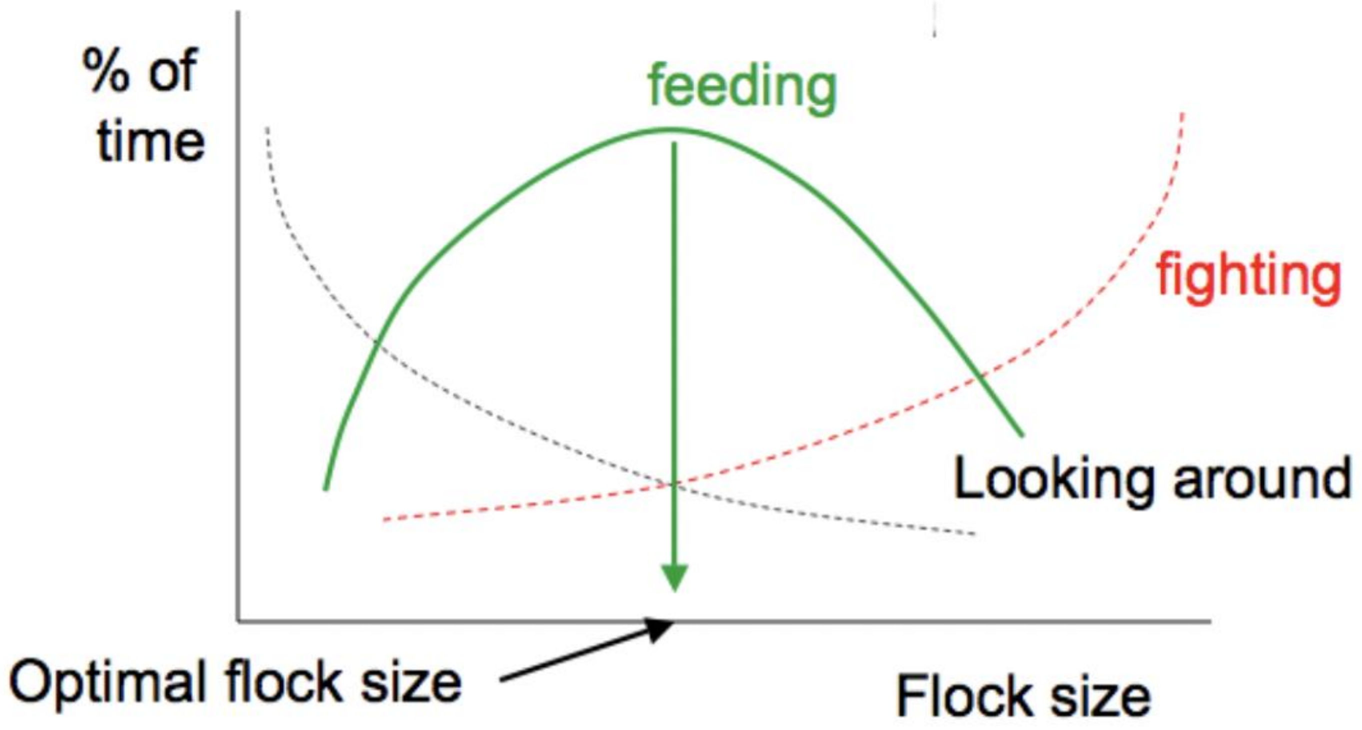




At what flock size can the bird spend the most time feeding (minimizing time spent fighting or looking around)?



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What if data do not match predictions?

- Context-dependent effects – do you need to account for phenotypic variation (sex, body size, etc.) or habitat differences?
- Interactions with other tasks – i.e., foraging choices might be different depending on extent to which individual has achieved other necessary tasks (rest, reproduction, social interactions, etc.)
- Used the wrong currency – maybe instead of time need to look at energy expended to perform behaviors

Optimal foraging theory

- Patch and prey models
- Two main types of questions:
  - Where to forage? Patch choice
  - What to eat? Prey choice

Handling time

- Certain prey items are harder to eat than others
- Crabs foraging in the intertidal sometimes do not eat the mussels they've dug up – why?

Handling time

- Certain prey items are harder to eat than others
- Crabs foraging in the intertidal sometimes do not eat the mussels they've dug up – why?
  - When a crab encounters a mussel, should it open it or should it search for something else?
  - Hypothesis: Eat items that lead to the highest gain in energy per unit time

How large a prey item should a predator attempt to eat, considering handling time and caloric content?

Energy/time

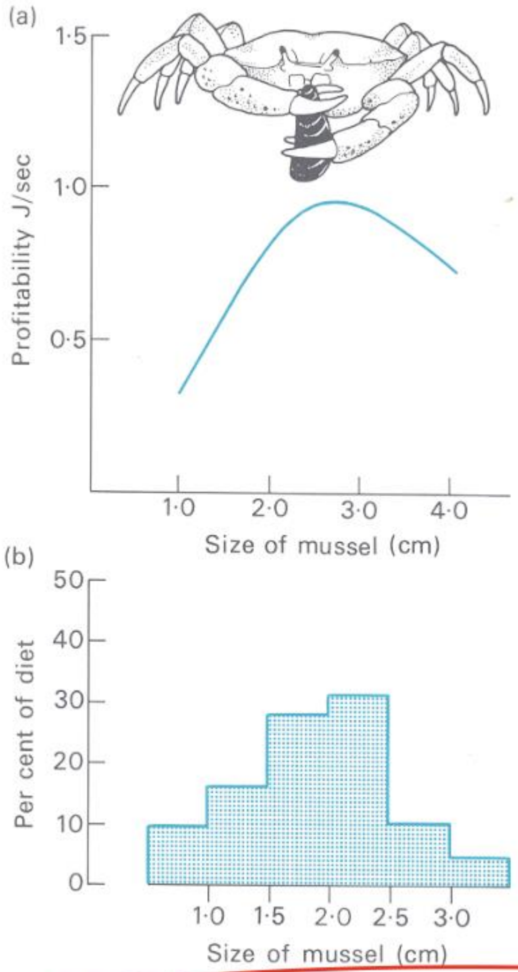

Fig. 3.5 Shore crabs (*Carcinus maenas*) prefer to eat the size of mussel which gives the highest rate of energy return. (a) The curve shows the calorie yield per second of time used by the crab in breaking open the shell and (b) the histogram shows the sizes eaten by crabs when offered a choice of equal numbers of each size in an aquarium. From Elner and Hughes (1978).

Charnov's optimal diet model

- Choice between big and small prey that vary in energy value (E) and handling time (h)
- Profitability of each prey type is E/h
- Assume large prey more profitable ($E_b/h_b > E_s/h_s$)
- How should predator maximize gain?
  - Encounter big prey = always eat
  - Encounter small prey = eat it, or keep looking?

Encounter small prey – eat it or keep looking?

- Need to consider the <u>search time (S)</u>
- Should eat small prey if gain from small prey is greater than the gain from large prey minus energy wasted searching for large prey
  - $E_s/h_s > E_b/(h_b + S_b)$
- The choice of the less profitable prey depends on the abundance of the more profitable prey


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ONE WAY

Example problem

- A badger hunts for either scorpions or ground squirrels, or both:
  - Scorpions provide 10 calories each, require 2 minutes to find an additional 3 min to remove the stinger
  - Ground squirrels offer 1000 calories each, require 3 hours to find and an additional 90 minutes to capture and consume.
- Our badger has encountered a squirrel: should it eat the squirrel? Now, assume that the badger has encountered a scorpion: should it eat the scorpion?

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$$\boxed{E_s/h_s > E_b/(h_b + S_b)}$$

$$\boxed{E_b/h_b > E_s/(h_s + S_s)}$$

Example problem

- Encounter squirrel first
  - 1000/90 > 10/(3+2)?
    - 11.1 > 2
- Encounter scorpion first
  - 10/3 > 1000/(90+180)?
    - 3.3 > 3.7

$$\boxed{E_s/h_s > E_b/(h_b + S_b)}$$

$$\boxed{E_b/h_b > E_s/(h_s + S_s)}$$

Ideal free distribution

- Trade-off between habitat quality and competition

"Ideal" – perfect information
"Free" – no exclusion
Replenishing food

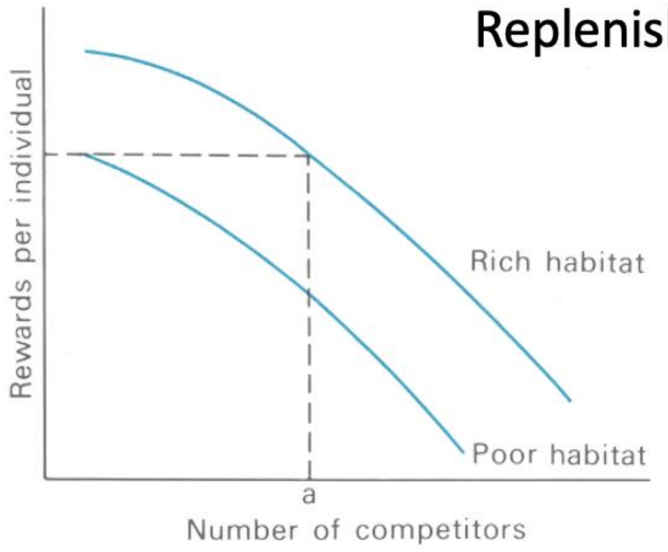
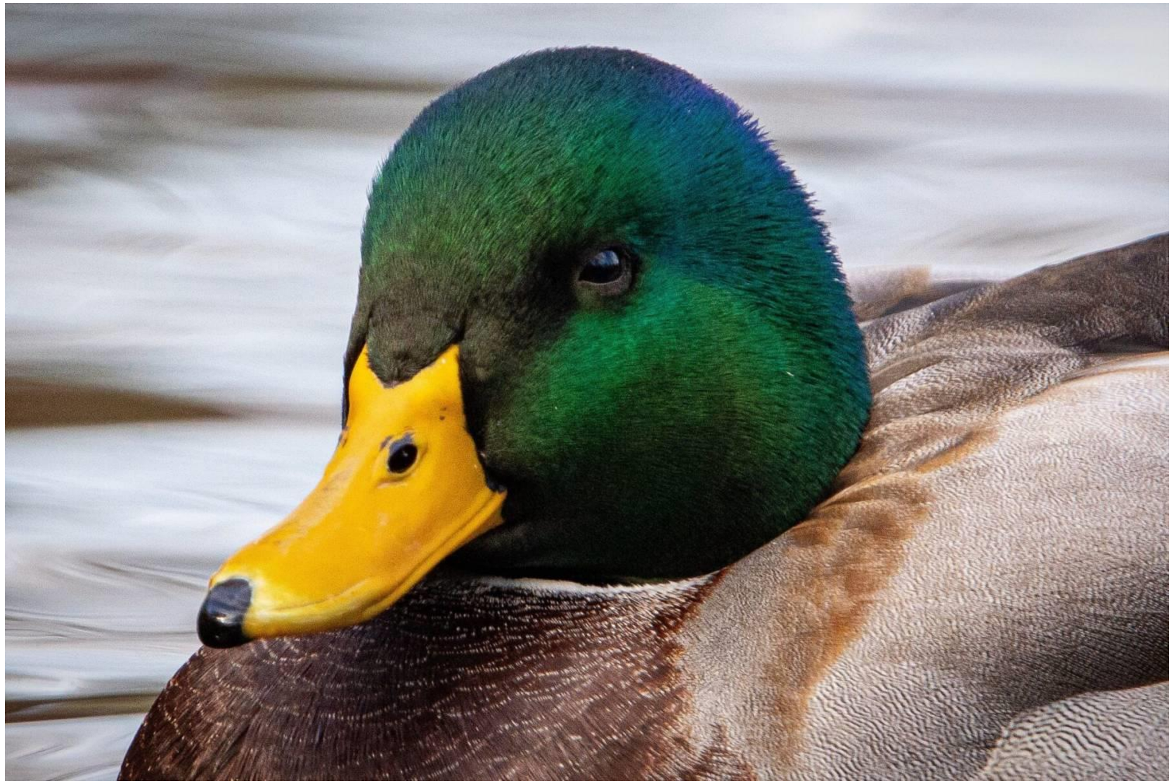


Fig. 5.1 The ideal free distribution. There is no limit to the number of competitors that can exploit the resource. Every individual is free to choose where to go. The first arrivals will go to the rich habitat. Because of resource depletion, the more competitors the lower the rewards per individual so at point *a* the poor habitat will be equally attractive. Thereafter the habitats should be filled so that the rewards per individual are the same in both. After Fretwell (1972).

Ideal free distribution in ducks

- 33 free-living mallards at Cambridge University
- Bread cubes thrown by two observers at fixed points on the lake 20m apart
- Patch profitability was varied by changing the rate at which bread was thrown



Harper 1982

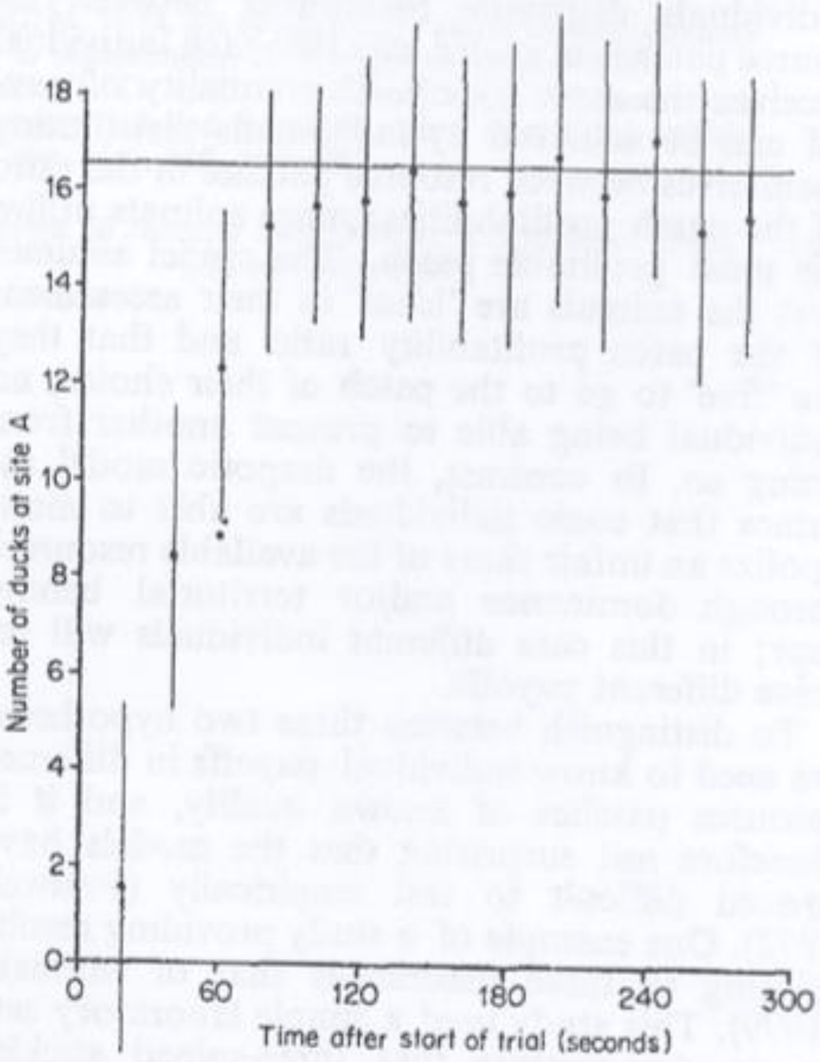


Fig. 1. Mean number of ducks at site A plotted against time since start of trial, when patch profitability ratio was unity. The horizontal line is the ideal free prediction.

Harper 1982

What's missing from the model?

Table II. Food Monopolization by the Six Most Successful Members of a Flock of 33 Ducks Competing at a Food Patch

Bird	Number of food items eaten in each trial											% Available food eaten in all trials
	1	2	3	4	5	6	7	8	9	10	11	
A	11	9	16	21	12	14	14	10	10	9	12	11.5
B	8	12	18	11	11	8	22	11	10	8	13	11.0
C	14	7	9	9	8	19	11	12	7	12	12	10.0
D	10	12	8	12	11	12	9	6	15	12	9	9.6
E	12	11	18	8	9	9	5	9	10	9	7	8.9
F	7	8	18	13	11	10	7	6	4	7	7	8.1
Total A to F	62	59	87	74	62	72	68	54	56	57	60	59.1
Total flock	111	107	102	114	119	100	121	117	100	104	109	100.0

Harper 1982

Critiques / limitations of optimality

- Animals don’t always have perfect information
- Assumes equilibrium
- Doesn’t account for learning/experiences
- Doesn’t account for any physiological, morphological, or social restrictions on behavior
- No individual variation

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2025 - Fall

- **Home**
- Announcements
- Assignments
- Grades
- Discussions
- Badges
- Office 365
- Course Gallery - Kaltura at Illinois
- Day1Access Course Materials

Proposal

Proposal Instructions

Choose a study to follow up and gather references for your proposal
Sep 16 30 pts

rubric for grading proposal - Fall 2025.xlsx

Sample proposal #1: Amato

Sample proposal #2: Laskowski

Sample proposal #3: Stein

Sample proposal #4: Neumann

Proposal

- Propose a research project that follows up on another study
  - Imagine you have all the resources possible at your disposal (within reason)
- Structure:
  - Introduction, Overview of experiment, Methods, Predicted Results, Significance and Broader impacts

Proposal timeline

Assignment	Due date	Points
Choose paper to follow up + 5-10 references	September 16th	30
Brainstorming	September 25th	30
Hypotheses and predictions	October 7th	40
Rough draft of methods	October 23rd	40
First draft	November 6th	--
Evaluation of partner's first draft	November 11th	60 (for first draft + evaluation of partner's first draft)
Second draft	November 20th	100
Final draft	December 12th	100

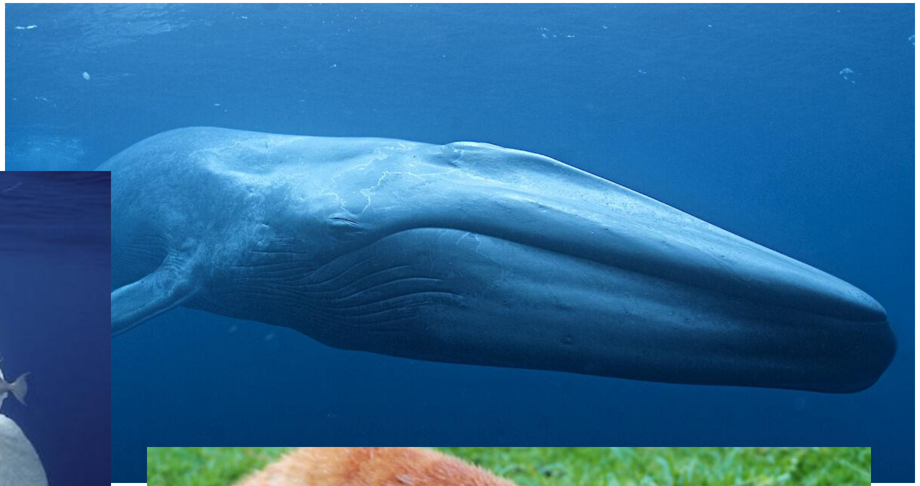
First assignment: choose a paper to follow up + gather references

1) Identify a paper to follow up
2) Gather 5-10 relevant references
3) Identify knowledge gap based on references
4) Present an idea for a question that would address this research gap (does not need to be what you end up doing, just get an idea on the page)

What does it mean to “follow up” a paper?

- If you would like, you can directly follow up on another study, using the same system and addressing gaps in the research left behind by the study
- If your ideas don't directly follow up on another paper; i.e., could be a gap in research that emerged from multiple papers, idea that is hinted at in the study but not directly referenced, then feel free to do that
- Finding paper to follow up is to provide you with some structure, not limit you

Write about something you are interested in!













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